Project Atharvam:

Technical Journey and Sprint Execution Report

Focus Area: Circular Economy and Green Synthesis of Activated Carbon

Startup Sprint Competition 2026

Timeline: February 4 – March 9, 2026

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1 Introduction

This report serves as the official documentation of the Atharvam startup. While the previously submitted Pitch Deck focused on market size and investment potential, this report provides a granular look at the Sprint Journey, the Technical Architecture of the synthesis process, and the Operational Milestones achieved between February 4 and March 9, 2026.

1.1 Energy Storage and Supercapacitors

During the initial literature review, we examined the role of Electric Double-Layer Capacitors (EDLCs), or supercapacitors, in modern energy storage systems. These devices store energy through electrostatic charge accumulation at the electrode–electrolyte interface, enabling rapid charge–discharge cycles and long operational lifetimes. Their performance is strongly governed by the structural properties of the electrode material.

1.2 Importance of Activated Carbon Electrodes

The investigation identified activated carbon as the dominant electrode material in commercial EDLCs due to its high specific surface area and tunable pore structure. However, most industrial activated carbon is produced from fossil-derived precursors using chemical activating agents such as KOH, indicating a reliance on energy-intensive and non-renewable production pathways.

1.3 Circular Economy Approach in Materials

This observation directed our attention toward biomass-derived carbon materials. Agricultural residues such as tea processing waste and banana plant biomass contain lignocellulosic structures suitable for carbonization and activation. Exploring these waste streams suggested the possibility of a localized circular pathway for producing activated carbon.

The core objective of this journey was to validate the transition from a "Linear Ma- terial Failure" (import-heavy carbon) to a "Circular Resource Model" (locally sourced agricultural waste).

2 The Linear Material Failure: A Deep Dive

The energy storage market in India is currently throttled by a supply chain paradox. The transition to Electric Vehicles (EVs) and renewable-heavy power grids relies on Electric Double-Layer Capacitors (EDLCs), which require high-surface-area activated carbon.

2.1 The Import Dependency Analysis

In 2024, India's import of activated carbon exceeded $98.2 million. Most of this material is derived from fossil-fuel-intensive processes in foreign markets. This creates:

- **Economic Vulnerability:** Exposure to global price fluctuations and shipping delays.
- **Sustainability Gap:** A clean energy device (the EV battery) powered by a carbon-heavy manufacturing process.

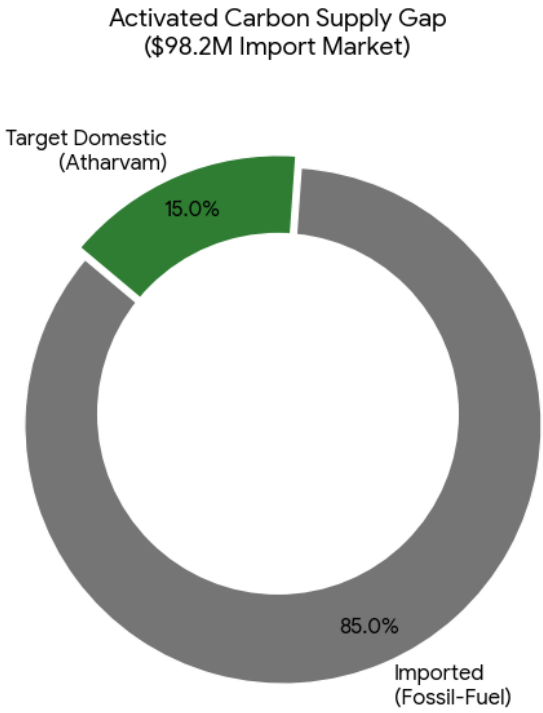

Figure 1: Activated carbon import dependency in India.

2.2 The Methane Crisis in Agriculture

Simultaneously, India's agricultural sector generates over 500 million tonnes of biomass. In the North-East, tea and banana residues are often left to undergo anaerobic decomposition.

$$\text{Biomass} \xrightarrow{\text{Anaerobic}} \text{CH}_4 + \text{CO}_2$$

Methane (CH_4) is significantly more damaging to the climate than CO_2. In some cases, biomass residues are diverted to **biogas production**, but this pathway often provides **limited economic returns for farmers**, as small-scale digesters produce low-value energy while requiring continuous feedstock management and infrastructure maintenance. As a result, much of this biomass remains untreated or poorly utilized.

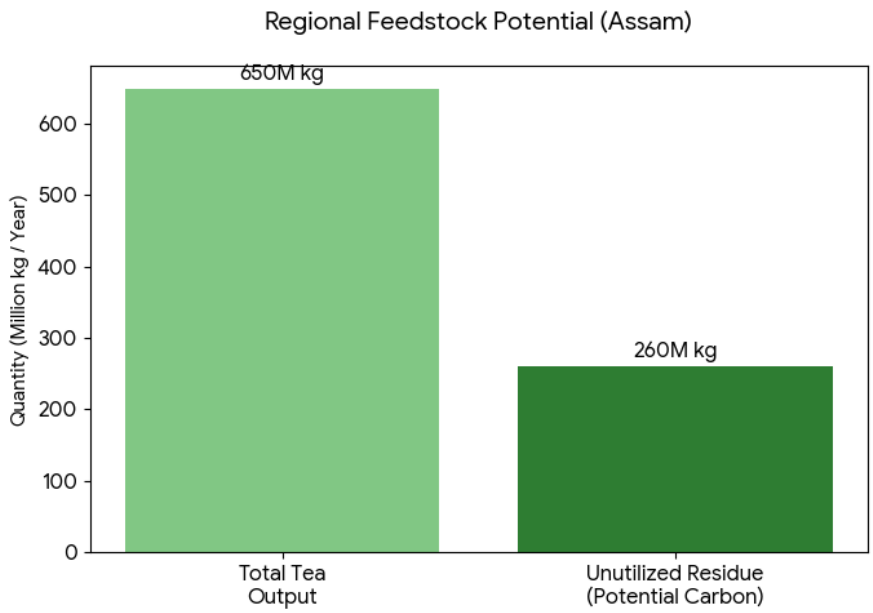

Figure 2: Tea biomass and residue potential in Assam.

By ignoring this waste, we are missing a feedstock that is naturally rich in the hierarchical structures required for **high-performance carbon electrodes**.

2.3 Environmental Cost of Conventional Activated Carbon

A review of conventional production pathways shows that most commercial activated carbon is derived from fossil-based precursors such as coal or petroleum coke and activated using strong chemical agents at high temperatures. While effective for

creating high surface area materials, this approach introduces several environmental and operational drawbacks.

- **Chemical and Energy Intensity:** Activation typically requires reagents such as KOH and temperatures above 700°C, resulting in high energy consumption and hazardous chemical handling.
- **Process Emissions:** The activation and washing stages generate chemical effluents and increase the overall carbon footprint of the material.

3 The Sprint Journey: Chronological Execution

The sprint was structured into four distinct phases, navigating through academic constraints and technical validation requirements.

3.1 Phase 1: Discovery and Resource Mapping (Feb 4 – Feb 10)

The first seven days focused on regional resource density. We mapped the concentration of tea estates in Assam (producing > 650 million kg of tea annually) and the corresponding waste volumes.

- **Objective:** To confirm if the "waste" volumes justify a decentralized refinery.
- **Outcome:** Confirmed that 30–40% of processed tea material is rejected as low-grade dust, providing a massive, steady supply of carbon precursors.

3.2 Phase 2: Technical Protocol Design (Feb 11 – Feb 20)

This phase focused on identifying a sustainable activation pathway that could integrate regionally available waste streams while eliminating synthetic chemical activators.

- **Objective:** Identify a compatible activating agent for tea-derived carbon using locally available biomass resources.
- **Innovation:** During the review of agricultural residues in Assam, banana plant waste was identified as a potential source of potassium-rich ash. Literature review subsequently revealed an approach described in *Indian Patent No. 377645* that utilizes banana plant ash extract ("khar") as a natural activating agent for tea-derived activated carbon.

3.3 Phase 3: The Strategic Hiatus (Feb 21 – March 6)

Status: Development Paused (Mid-Semester Examinations). A critical component of this sprint journey was managing external constraints. For 14 days, active prototyping was suspended for academic priorities.

- **Internal Audit:** Used this time for "Passive Review" of existing literature on hierarchical carbon morphology.
- **Synthesis:** Refined the theoretical impregnation ratios of banana extract to tea powder.

3.4 Phase 4: High-Velocity Finalization (March 7 – March 9)

Post-examination, the sprint resumed with a focus on documenting the operational model and finalizing this journey report. The final phase focused on consolidating the technical framework developed during earlier stages and translating it into a structured operational model suitable for presentation and further validation.

4 Technical Methodology: Green Synthesis

This section details the proposed manufacturing process derived during the sprint.

4.1 Bio-Chemical Activation Mechanism

In traditional activation, KOH creates pores by the following simplified reaction at high temperatures:

$$6\mathrm{KOH} + \mathrm{C} \rightarrow 2\mathrm{K} + 3\mathrm{H_2} + 2\mathrm{K_2CO_3}$$

Our process utilizes the natural potassium carbonate (K_2CO_3) found in banana plant ash. This avoids the use of pure KOH, which is corrosive and hazardous to handle.

4.2 The Five-Step Process Flow

1. **Feedstock Stabilization:** Tea processing waste is collected, washed, and dried to a constant weight to remove moisture and surface silt.

2. **Activating Agent Extraction:** Banana stems are incinerated in a controlled environment to produce ash. This ash is leached in water to create a concentrated potassium-rich extract.

3. **Impregnation:** The dried tea waste is soaked in the extract. This ensures the activator is distributed at the cellular level within the hierarchical tea leaves.

4. **Pyrolysis/Carbonization:** The material is heated to temperatures between 600°C and 800°C in an inert atmosphere.

5. **Refining:** Aqueous washing removes any residual mineral salts, leaving behind a highly porous carbon framework.

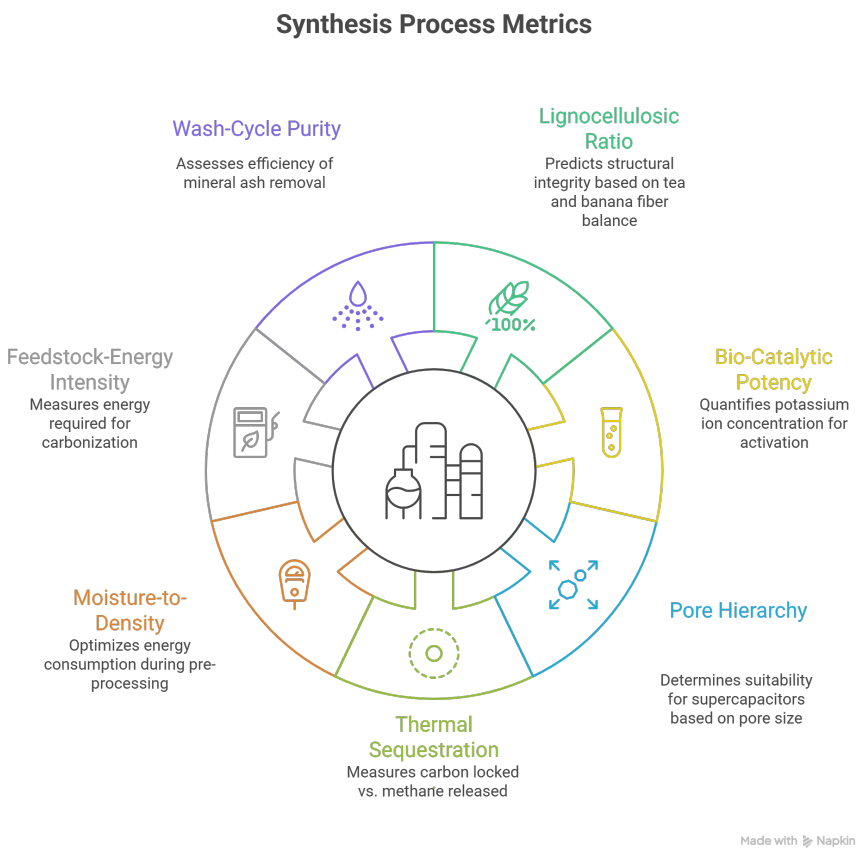

Figure 3: Key parameters in the carbon synthesis process.

4.3 Utilizing Existing Laboratory Facilities at IIT Guwahati

Laboratory access required for the proposed synthesis can be facilitated through the Technology Incubation Centre (TIC) at IIT Guwahati. This enables small-scale validation of the process steps.

- **Feedstock Stabilization:** Laboratory drying ovens available in materials laboratory can be used to remove moisture from collected tea residues prior to processing.
- **Pre-processing and Size Reduction:** Standard grinding equipment and sieve systems available in shared laboratory spaces can be used to obtain uniform biomass particle sizes required for consistent activation.
- **Activator Preparation:** Small-scale controlled combustion setups can be used to convert banana plant residues into ash, followed by aqueous extraction and filtration using standard laboratory glassware.
- **Carbonization:** High-temperature furnaces available in existing laboratories can be used to conduct pyrolysis in the range of $600°C$ to $800°C$ under controlled atmospheric conditions.
- **Refining and Post-processing:** Filtration systems, washing setups, and drying ovens can be used to remove residual salts and obtain the final porous carbon material.

Category	Allocation	Amount	Key Expenditure Items
Essential R&D & Lab	55%	₹ 4,12,500	Small-scale muffle furnace, chemical glassware, and outsourcing advanced testing (CV/GCD) to university labs.
Feedstock & Logistics	15%	₹ 1,12,500	Manual collection from 2 local tea estates; basic solar-drying setup for biomass stabilization.
Ops & Intern Stipends	20%	₹ 1,50,000	2 Part-time research interns; founder-led execution.
Provisional IP & Misc	10%	₹ 75,000	Provisional patent filing and basic material characterization fees.

Figure 4: Proposed allocation of startup resources.

5 Logistical and Regional Integration

A startup is more than its chemistry; it is defined by the robustness of its supply chain. During the sprint, we developed a preliminary model for a "North-East Bio-Refinery Network" that connects agricultural waste streams with carbon material production.

5.1 Decentralized Collection Model

Transporting raw biomass is inefficient due to its high volume-to-weight ratio. To address this constraint, the proposed model separates early-stage processing from final carbon synthesis.

- **Satellite Pre-processing Units:** Small-scale drying and grinding facilities located near tea clusters in Assam can reduce biomass volume prior to transport.
- **Central Refinery:** A centralized facility would perform high-temperature carbonization and post-processing steps in order to maintain material consistency and quality control.

5.2 Competitive Advantage Analysis

The proposed Atharvam model differs from conventional activated carbon supply chains in both feedstock selection and activation strategy. Whereas traditional production relies on fossil-derived precursors and synthetic activating agents, the proposed approach utilizes agricultural waste and naturally occurring alkaline compounds.

Key distinctions include:

- **Feedstock Source:** Conventional activated carbon typically uses coal or petroleum-derived materials, whereas the proposed model utilizes tea processing waste.
- **Activation Pathway:** Instead of synthetic KOH activation, the process employs potassium-rich banana plant ash extract.
- **Supply Chain Structure:** The use of regional biomass reduces import dependence and enables a localized circular material pathway.

5.3 Economic Feasibility Discussion

A preliminary economic assessment suggests that utilizing locally available biomass residues can reduce the overall cost of activated carbon production.

- **Low-cost Precursors:** Agricultural residues such as tea dust and banana plant waste are inexpensive and widely available.
- **Reduced Transportation Costs:** Decentralized preprocessing minimizes the need to transport bulky raw biomass over long distances.
- **Additional Rural Value Streams:** Converting agricultural residues into industrial feedstock creates potential supplementary income for plantation operators and farmers.

5.4 Regional Impact in North-East India

Beyond technical feasibility, the proposed system may contribute to broader regional development by integrating agricultural waste streams into a value-added materials supply chain.

- **Employment Opportunities:** Satellite preprocessing units and centralized refining facilities could generate technical and operational employment.
- **Waste Utilization:** Large quantities of rejected tea dust produced during processing can be redirected into carbon material production.
- **Localized Industrial Development:** Linking agricultural resources with advanced carbon materials may support the emergence of a clean-materials ecosystem within the North-East region.

6 Lessons Learned and Future Roadmap

The sprint journey highlighted the importance of **context-aware innovation**.

Feature	Traditional Imports	Atharvam Model
Precursor	Coal/Fossil Fuel	Agricultural Waste
Activator	Synthetic KOH	Banana Plant Extract
Environmental Impact	High CO_2	Negative Methane
Logistics Cost	High (Import Duties)	Low (Regional Sourcing)
Resource Type	Linear	Circular

Table 1: Comparison of traditional vs. proposed synthesis models.

6.1 The "Midsem" Insight

The forced pause during mid-semester exams allowed us to realize that the most complex part of the startup isn't the chemistry, but the seasonal availability of banana waste versus the year-round supply of tea dust. This realization informed a revised "Seasonal Stockpiling" strategy.

6.2 Future Milestones

Following the conclusion of this sprint on March 9:

- **Q2 2026:** Laboratory optimization of the SSA (Specific Surface Area) targeting $> 1500\ \mathrm{m}^2/\mathrm{g}$.
- **Q3 2026:** Pilot partnership with a regional tea estate for onsite waste collection.
- **Q4 2026:** Prototype testing in a 10F supercapacitor cell.

7 Conclusion

Project Atharvam has evolved from a theoretical concept into a validated operational roadmap. Despite the academic pressures and the logistical complexity of the North-East, the sprint has proven that the solution to India's energy storage dependency is currently being discarded as waste. By linking the tea industry to the EV industry, we can turn a $98.2 million economic drain into a sustainable, localized industrial powerhouse.

8 Photo Gallery

Figure 5: Team working on the startup side by side in the hostel rooms and common areas.

Figure 6: Meeting a farmer living near Khokha.